

# **EECS 498/CSE 598: Zero-Knowledge Proofs**

## **Winter 2026**

### **Lecture 22**

Paul Grubbs

[paulgrub@umich.edu](mailto:paulgrub@umich.edu)

Beyster 4709

# Agenda for this lecture

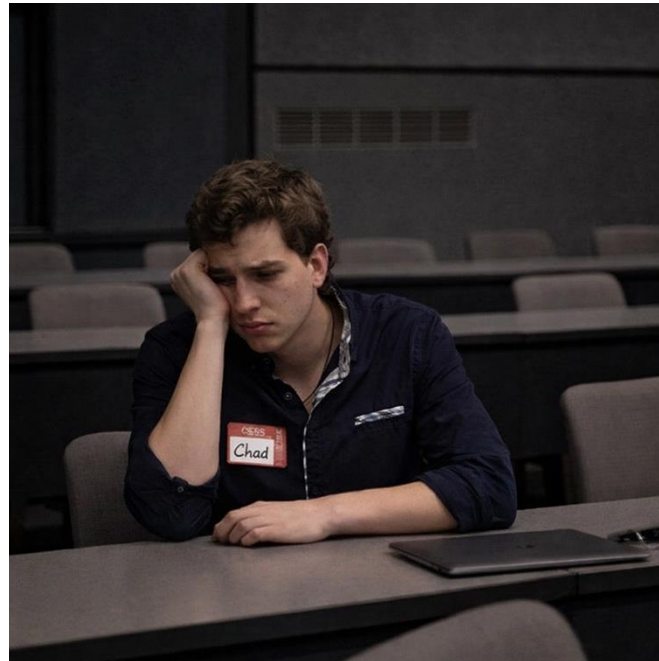
- Announcements
- Background on ecash and blockchains
- Sander-Ta Shma
- Zerocash
- Other applications
  - zkVMs, rollups
  - private smart contracts

# Agenda for this lecture

- **Announcements**
- Background on ecash and blockchains
- Sander-Ta Shma
- Zerocash
- Other applications
  - zkVMs, rollups
  - private smart contracts

# Announcements

- P4 and theory pset 2 are out!.
  - Note there will be no P5 nor a final exam. These will be the last assignments of the semester.
- All work for this course must be completed by our final exam date, April 27<sup>th</sup>
- Reminder: you can get participation credit for going to discussion.
  - Chad is getting lonely on Fridays!

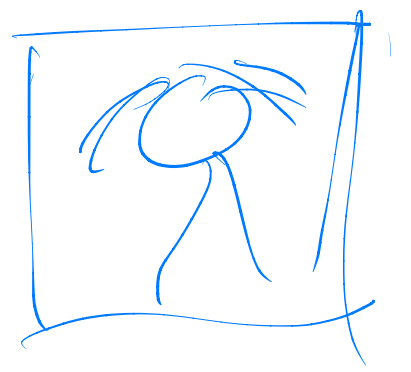


# Agenda for this lecture

- Announcements
- Background on ecash and blockchains
- Sander-Ta Shma
- Zerocash
- Other applications
  - zkVMs, rollups
  - private smart contracts

# Chaumian ecash

- Design basic digital payments
  - Can't lie about amounts of money
  - Payment privacy



Chaum's proposal:

1. Withdraw money using blind sig

User                      Bank

SN      blinded sn →       $\sigma \leftarrow \text{Sign}(\text{blinded sn})$

denote ←  $\sigma$

$\sigma_{\text{SN}}$

2. To spend at merchant
  - Show SN,  $\sigma_{\text{SN}}$  to merchant
  - Merchant asks bank to check for double-spends of SN

# Blockchains

- distributed ledger

consistent view of actions

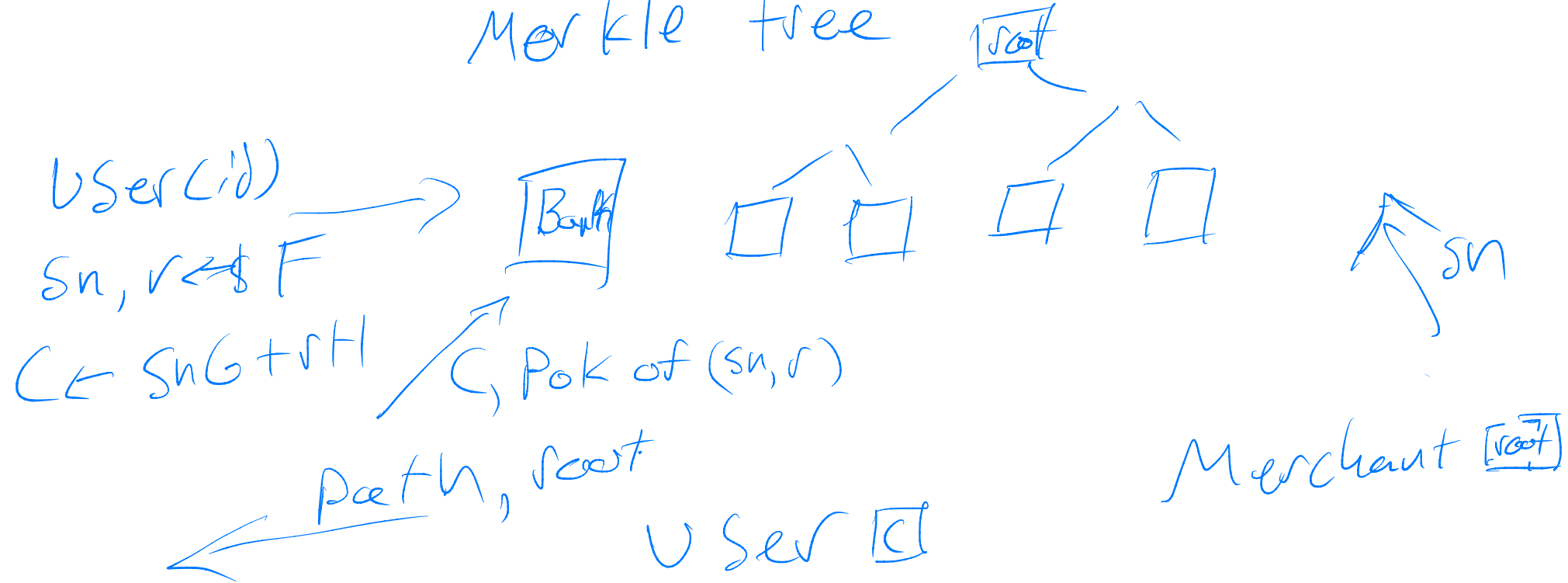
provides "consensus" on system state

# Agenda for this lecture

- Announcements
- Background on ecash and blockchains
- **Sander-Ta Shma**
- Zerocash
- Other applications
  - zkVMs, rollups
  - private smart contracts

# Sander-Ta Shma

Offline bank: put all issued coins in  
Merkle tree



# Sander-Ta Shma

Extends to double-spend

During coin creation user generates  $V_1 \in \mathbb{Z}$   
 $V_2 = V_1 - id$ ,  $C = V_1 G_1 + V_2 G_2 + sn G_3 + \sigma H$

User proves knowledge of repr  $(V_1, V_2, sn, \sigma)$

and  $V_1 - V_2 = id$

During spend,  $chal \leftarrow H(\text{merchant ID}, \text{timestamp})$

Reveal  $V$  and  $sn$  for merchant, PoK of  $C$  and path,

and  $V_1 = V_1 + chal_1 \cdot V_2$

IF Double spend:  $V_2 = V_1 + chal_2 \cdot V_2$

# Agenda for this lecture

- Announcements
- Background on ecash and blockchains
- Sander-Ta Shma
- Zerocash 2012
- Other applications
  - zkVMs, rollups
  - private smart contracts

# address secret key **Zerocash**

USER  $a_{sk}$  address public key  $a_{pk} = \text{PRF}_{a_{sk}}(0)$

A coin is a tuple  $(a_{pk}, v, p, r, s, cm)$

•  $v$  is value of coin

•  $p$  serial no randomness

•  $r, s$  is com randomness

• Coin commitment is

$$k = \text{COM}_r(a_{pk} || p), \quad cm = \text{COM}_s(v || k)$$

Ledger has MT over all coin commits and list of spent SAs  
Coin "Mint" value  $v$  for  $a_{pk}$ ,

Sample  $p, r, s$ , Compute  $k = \text{COM}_r(a_{pk} || p)$  and  $cm = \text{COM}_s(v || k)$

Post  $tx_{\text{mint}} = (cm, v, k, s)$  to blockchain  
+ PoK of  $a_{pk}$

# Zerocash

[invariant:  $v_{old} = v_1^{new} + v_2^{new}$ ]

"Pour"

Start with  $C_{old}$

, pour  $C_1^{new}, C_2^{new}$

to  $a_{PK,1}^{new}$  and  $a_{PK,2}^{new}$

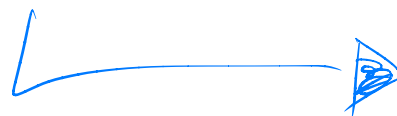
1. Compute  $SN = PRF_{sk}^{old}(P_{old})$

2. Sample  $P_1^{new}, r_1^{new}, s_1^{new}$   $P_2^{new}, r_2^{new}, s_2^{new}$

Compute  $k_1^{new}, k_2^{new}$

$CM_1^{new}$   
 $CM_2^{new}$

3. Generate ZKP for statement



# Zerocash

public inputs

# Agenda for this lecture

- Announcements
- Background on ecash and blockchains
- Sander-Ta Shma
- Zerocash
- Other applications
  - zkVMs, rollups
  - private smart contracts

# Other ZKP applications